Lead Service Line Infrastructure Needs Across U.S. States - Executive Summary

Project Context

This project uses the U.S. EPA's 2026 Q2 Service Line Inventory to examine where lead service line infrastructure needs are greatest across U.S. states. The analysis covers 43,993 Community Water Systems across the 50 states and District of Columbia that reported all required service-line types.

Rather than combining all service-line needs into one score, the analysis separates two different parts of the problem. Identification Need is measured by Lead Status Unknown service lines, while Known Replacement Inventory is measured by Lead plus Galvanized Requiring Replacement (GRR) lines. This distinction is important because an Unknown line is not assumed to be lead; it is a line whose material still needs to be classified.

Core Insight: One Ranking Hides Different Planning Needs

The most useful finding is that there is no single meaningful way to rank state lead service line need. Identification Need exceeds Known Replacement Inventory in 49 of 51 states/DC, and nationally the analytical sample contains 21.59 million Unknown lines compared with 3.12 million lines already classified for replacement.

Even with that broad pattern, the balance between the two needs varies sharply by state. Illinois has the largest Known Replacement Inventory at 672,210 lines, while Texas and Pennsylvania have the largest unresolved identification inventories at about 2.18 million and 2.06 million Unknown lines. Ohio and New York have substantial inventories on both dimensions. Only five states appear in both raw top-10 lists, showing that the largest identification and replacement workloads are not concentrated in the same places.

Because these measures represent different stages of infrastructure work, a high position in one ranking does not mean the same thing as a high position in the other. For Unknown lines, material must first be classified before replacement need can be determined. Known Lead and GRR lines already represent defined replacement work that can move into planning. These activities can happen at the same time; replacement planning for known lines does not have to wait until every Unknown line is resolved.

Population Context Adds a Second Lens

Raw totals answer one planning question: where are the largest absolute inventories? However, raw counts also tend to favor larger states, so 2025 Census resident population estimates were used as a size-adjustment benchmark.

The normalized results answer a different question: where is the inventory large relative to state population? Only 6 of the raw top 10 Known Replacement states and 3 of the raw top 10 Identification Need states remain in their respective top 10 after normalization. Illinois remains highest for Known Replacement at 528.5 per 10,000 residents, while Pennsylvania leads Unknown inventory at about 1,574 per 10,000 residents.

This shows why state need should be viewed through more than one lens. Raw totals help identify the largest overall workloads, while population-adjusted values help reveal where inventories are especially large relative to state size. These rates provide context only; they do not estimate the number of affected residents or customers.

Implications and Next Question

Taken together, the results suggest that state lead service line need is better understood through several planning questions rather than one overall ranking. Absolute workload shows how much inventory is involved, Identification Need shows how much of that inventory still lacks material certainty, and population adjustment shows where reported inventory is especially large relative to state size.

As more Unknown lines are classified, the replacement workload becomes clearer. At the same time, already-known Lead and GRR inventories provide a more defined starting point for replacement

planning. The state-level analysis can show where deeper investigation may be valuable, but actual intervention decisions still depend on utility-level conditions, geography, funding, local capacity, inspections, and project-level information.

An important limitation is that Unknown does not mean lead. Some Unknown lines may ultimately be classified as non-lead, so the 21.59 million Unknown lines should not be interpreted as 21.59 million lines requiring replacement. The inventory also does not measure household lead exposure or water quality.

Recommended Next Question

Once service lines are identified, which utility-level, geographic, funding, and local-capacity factors determine where replacement can move forward?